



Correspondence

REPLY: We thank Dr Sabour and Dr Ghassemi for their comments regarding our paper.¹ The authors expressed 2 concerns about our methodology and statistical analysis.



Regarding the gradeability concordance, we evaluated all 1969 images that were each graded by both primary readers (Table 1) and used their grades to calculate a prevalence and bias-adjusted kappa of 0.84. This value suggests very strong agreement between the 2 readers which further supports our conclusion of high interreader reproducibility of gradeability.²

With respect to the quantitative variables, we previously determined that the intraclass correlation coefficient (ICC) (2, 1)³ for lesion area was 0.99 (95% confidence interval also 0.99–0.99). We opted to report Bland-Altman and coefficient of reproducibility (smallest real difference) rather than the ICC, as ICC is known to be highly dependent on the range of values measured, with greater range leading to higher ICC.⁴ Finally, it is important to note that all measurements were calculated with an individual-based approach rather than the global average approach. Each image included in the lesion area analysis was graded by the same 2 readers and the interreader difference was calculated in a consistent manner.

ANGELA S. LI, MD

JUSTIN MYERS, BS

SANDRA S. STINNETT, DRPH

DILRAJ S. GREWAL, MD

GLENN J. JAFFE, MD

Department of Ophthalmology, Duke University, Durham, North Carolina

Disclosure(s):

All authors have completed and submitted the ICMJE disclosures form.

The authors have made the following disclosures:

D.S.G.: Consultant — Genentech, EyePoint, and Regeneron.

Table 1. Gradeability Data between 2 Primary Graders

Grader 1	Grader 2		Total
	Yes	No	
Yes	1722	71	1793
No	87	89	176
Total	1809	160	1969

This analysis was done for all 1969 images that were graded by the 2 primary readers, noted here as Grader 1 and Grader 2. Prevalence and bias-adjusted kappa was 0.84 based on this table.

G.J.F.: Consultant — Roche/Genentech, Annexon, 4DMT, Eyepoint, and Regeneron.

Available online: July 17, 2024.

Correspondence:

Glenn J. Jaffe, MD, Department of Ophthalmology, Duke University, Durham, NC 27710. E-mail: jaffe001@mc.duke.edu.

References

1. Li AS, Myers J, Stinnett SS, et al. Gradeability and reproducibility of geographic atrophy measurement in GATHER-1, a Phase II/III randomized interventional trial. *Ophthalmol Sci*. 2024;4:100383.
2. McHugh ML. Interrater reliability: the kappa statistic. *Biochem Med*. 2012;22:276–282.
3. Shrout PE, Fleiss JL. Intraclass correlations: uses in assessing rater reliability. *Psychol Bull*. 1979;86:420–428.
4. Patton N, Aslam T, Murray G. Statistical strategies to assess reliability in ophthalmology. *Eye (Lond)*. 2006;20:749–754.